



Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	2026-27 (ODD)	Course Name	LAB: Software Engineering
Semester	7 th	Course Code	23CT1704
Roll No	25	Name of Student	Tanushri Sawalikar

Practical Number	03
Aim	To study and draw the Data Flow Diagrams at Level 0 and Level 1 for the given case study of an E-Learning Platform.
Theory (100 words)	A Data Flow Diagram (DFD) is a graphical representation of the flow of data through an information system. It shows how data enters the system, how it is processed, where it is stored, and how it is transferred to external entities. A Level 0 DFD, also called a context diagram, represents the complete system as a single process and shows its interaction with external entities. A Level 1 DFD decomposes the main process into smaller functional processes and data stores. For an E-Learning Platform, DFDs represent registration, course management, learning content, assessment, payment, reporting, notifications, and interactions with students, instructors, and administrators.
Procedure and Execution (100 Words)	Algorithm(Steps): Step 1: Start. Step 2: Identify the external entities: • Student • Instructor • Administrator • Payment Gateway Step 3: Identify the main system: • E-Learning Platform Step 4: Draw the Level 0 DFD with the E-Learning Platform as process 0.0 . Step 5: Add data flows between the external entities and the E-Learning Platform. Step 6: Decompose the E-Learning Platform into Level 1 processes: 1.1 Registration Management 1.2 Course Management 1.3 Learning & Content Management 1.4 Assessment Management 1.5 Payment Management



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1.6 Report & Notification Management

Step 7: Add data stores:

- D1 – User Database
- D2 – Course Database
- D3 – Content Database
- D4 – Assessment Database
- D5 – Payment Database
- D6 – Notification Database

Step 8: Connect the processes with appropriate data flows.

Step 9: Verify the completeness and consistency of both DFD levels.

Step 10: Stop.

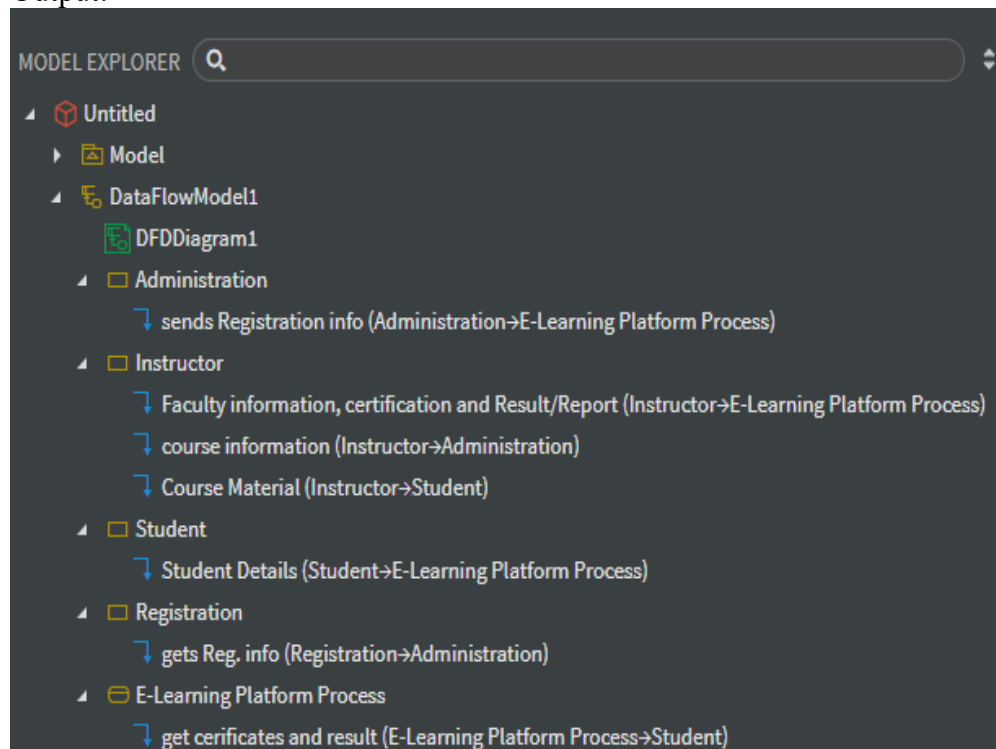
Implementation:

Tool used: StarUML

The diagrams can be created using standard DFD symbols:

- Rectangle → External Entity
- Rounded Rectangle/Circle → Process
- Open rectangle/parallel lines → Data Store
- Arrow → Data Flow

Output:





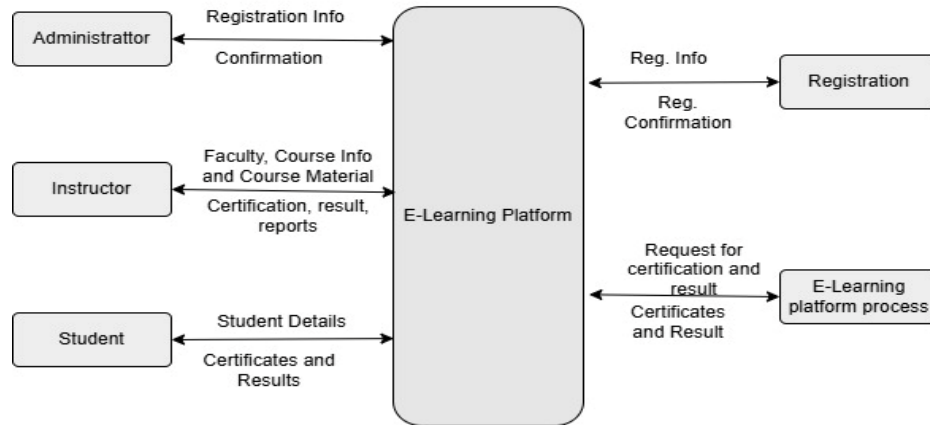
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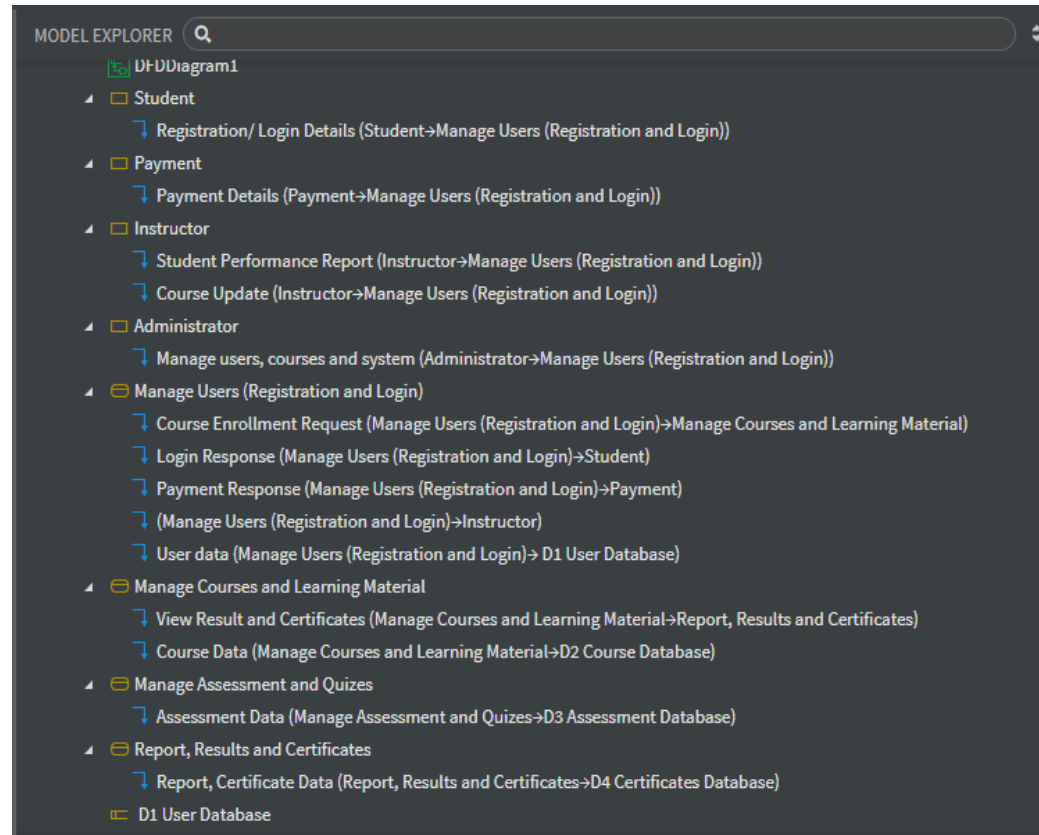
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Level0 Diagram for E-Learning Platform

Level1:



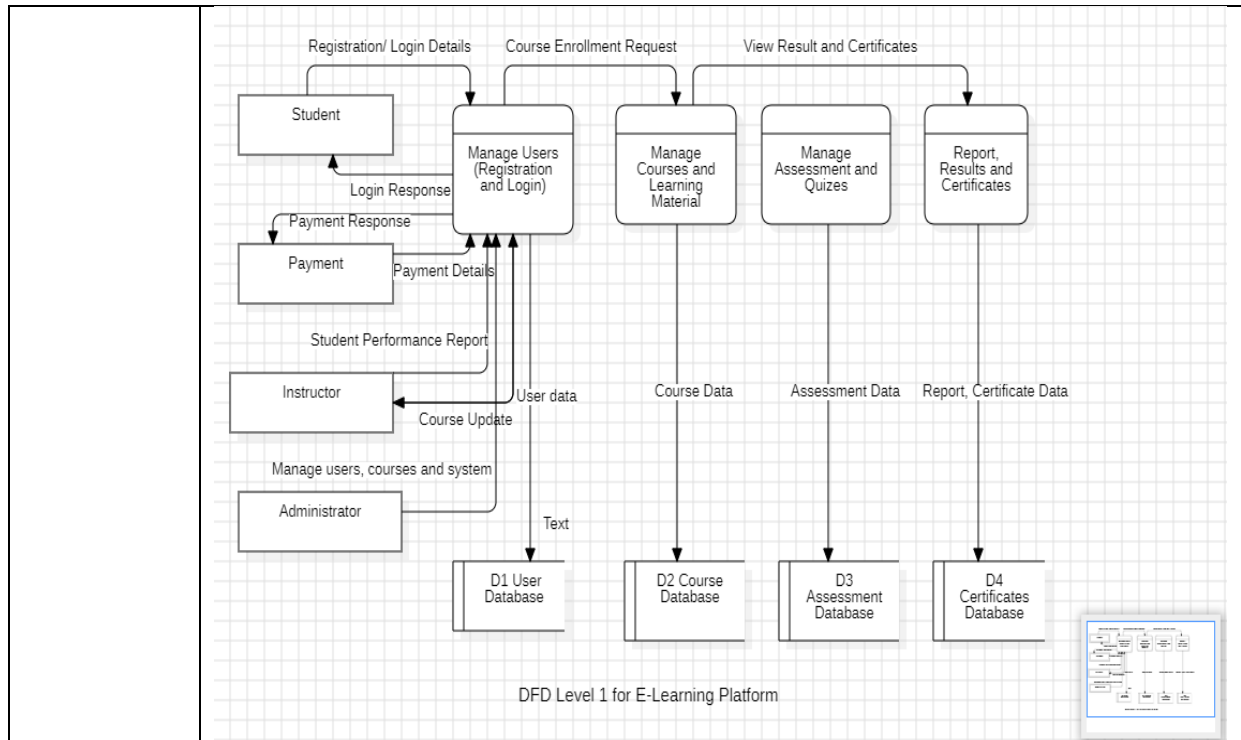
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Output Analysis	The Level 0 DFD provides a high-level view of the E-Learning Platform and identifies the major external entities interacting with the system. It shows how students, instructors, administrators, and the payment gateway exchange information with the platform. The Level 1 DFD provides a more detailed representation by decomposing the platform into registration, course, learning content, assessment, payment, and reporting/notification processes. The associated databases show how information is stored and retrieved. The DFDs provide a clear understanding of the movement and processing of data within the E-Learning Platform and help identify the functional requirements of the system.
Link of student Github profile where lab assignment has been uploaded	https://github.com/TanushriSawalikar/Software-Engineering-Practicals
Conclusion	I have successfully understand the concept of Data Flow Diagram at Level 0 and Level 1 for the case study E-learning Platform



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